

# Checkpoint

## 14.5: The Chain Rule

**Problem 1.** Write  $f(x) = \cos\left(\sqrt{5}e^{-7x}\right)$  as a composition of functions using different variables.

**Problem 2.** If  $f_x = 8xy$  and  $f_y = \tan^{-1}(x + y)$  what is  $\frac{dy}{dx}$ ?

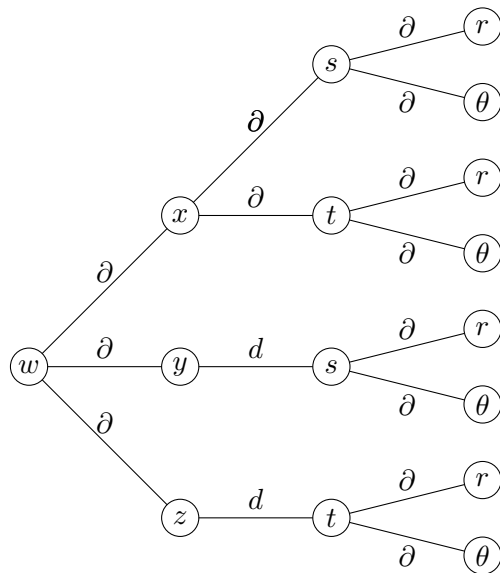
**Problem 3.** Draw the tree diagram for the following composition of functions and label the edges with partials/derivatives.  $w = f_1(x, y, z)$ ,  $x = f_2(s, t)$ ,  $y = f_3(s)$ ,  $z = f_4(t)$ ,  $s = f_5(r, \theta)$ , and  $t = f_6(r, \theta)$ .

**Problem 4.** In the above problem find an expression for  $\frac{\partial w}{\partial r}$ .

# 1 Answers

1.  $y = \cos u$ ,  $u = \sqrt{v}$ ,  $v = 5e^w$ ,  $w = -7x$ .

2.  $-8xy/\tan^{-1}(x+y)$ .



3.

4.  $\frac{\partial w}{\partial r} = \left(\frac{\partial w}{\partial x}\right)\left(\frac{\partial x}{\partial s}\right)\left(\frac{\partial s}{\partial r}\right) + \left(\frac{\partial w}{\partial x}\right)\left(\frac{\partial x}{\partial t}\right)\left(\frac{\partial t}{\partial r}\right) + \left(\frac{\partial w}{\partial y}\right)\left(\frac{dy}{ds}\right)\left(\frac{\partial s}{\partial r}\right) + \left(\frac{\partial w}{\partial z}\right)\left(\frac{dz}{dt}\right)\left(\frac{\partial t}{\partial r}\right)$